



VMware vSAN to Proxmox VE Ceph Migration

Case Study



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Executive Summary

Data Consultants Corporation (DCC), a US-based IT services organization, faced a critical infrastructure decision following Broadcom's acquisition of VMware. A sweeping change in VMware's licensing model — moving from perpetual to mandatory subscription-based pricing — resulted in a dramatic and unplanned escalation in infrastructure costs, directly threatening DCC's operational margins. Leadership made the strategic decision to migrate away from VMware entirely and adopt an open-source, enterprise-grade alternative, with two firm constraints: no new hardware procurement, and no application downtime exceeding 24 hours.

DCC engaged **Illumia Solutions Pvt. Ltd. (ISPL)**, a Proxmox Silver Partner, to lead the end-to-end migration from VMware vSphere with vSAN to **Proxmox VE with Ceph**, with **Proxmox Backup Server** replacing the existing Veeam Backup & Replication environment.

ISPL designed and executed a phased, non-disruptive migration strategy that worked entirely within DCC's existing hardware fleet. Rather than a high-risk big-bang cutover, ISPL employed an iterative approach — selectively freeing nodes from the VMware cluster, re-provisioning them as Proxmox VE nodes, and progressively migrating workloads across — cycling through the entire infrastructure until VMware was fully decommissioned. Proxmox VE's native ESXi-to-PVE migration tooling ensured that individual workloads experienced minimal disruption, well below the agreed downtime threshold.

The outcomes of the engagement were unambiguous:

- **VMware fully decommissioned** — all hosts re-provisioned as Proxmox VE nodes
- **All virtual machines successfully migrated** to the new Proxmox VE + Ceph cluster
- **Veeam replaced** with Proxmox Backup Server, with encryption at rest and a defined retention policy
- **No new hardware procured** — the entire migration executed on existing infrastructure
- **Licensing costs reduced drastically** — DCC's dependence on Broadcom's VMware subscription model eliminated completely

DCC now operates on an integrated, open-source infrastructure stack with no vendor lock-in, predictable costs, and a strengthened data protection posture — delivered on time, within budget, and without disruption to business operations.

Background Information

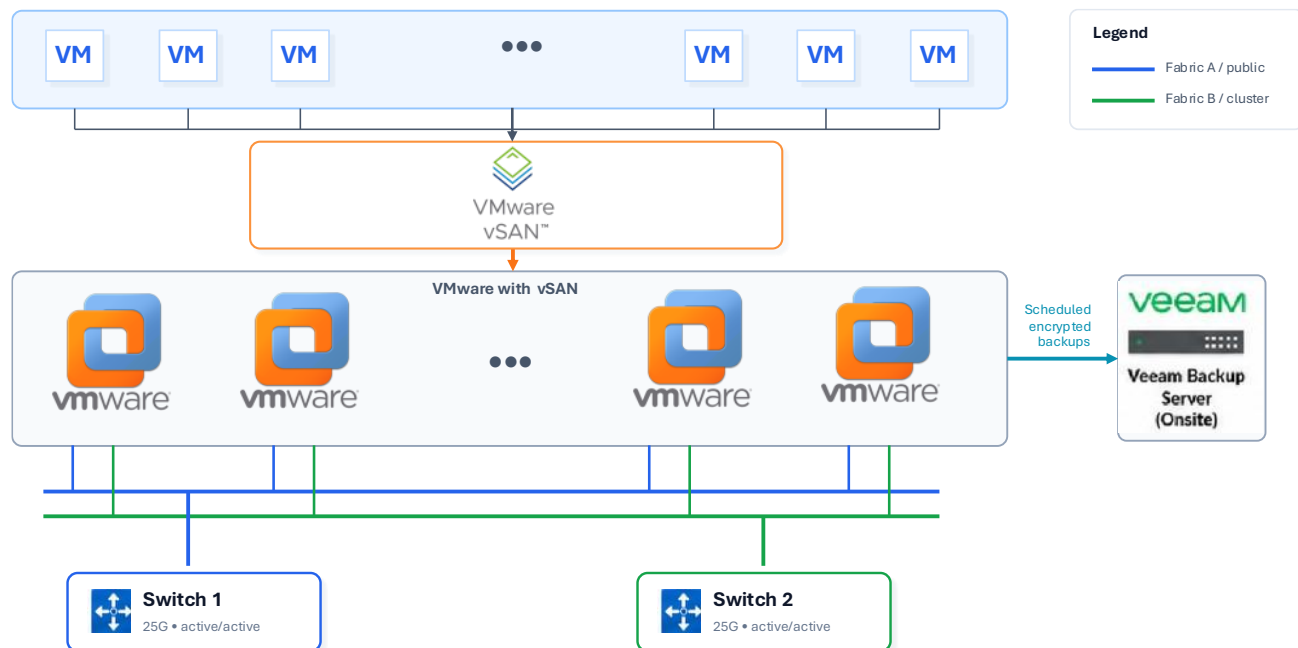
Data Consultants Corporation (DCC), a US-based IT services organization, had built its virtualization infrastructure on VMware vSphere with vSAN — a proven and widely adopted enterprise platform. The environment comprised multiple physical host nodes interconnected through a pair of high-throughput 25G storage switches, forming a robust hyper-converged cluster. Virtual machine backups were managed through Veeam Backup & Replication, providing reliable on-site data protection.

As an IT services company, DCC's infrastructure is not just an internal support function — it is the operational backbone on which client-facing services and internal workloads run. Maintaining performance, availability, and data integrity is therefore a business-critical requirement, not merely an IT concern.

In 2024, Broadcom's acquisition of VMware resulted in sweeping changes to VMware's licensing model. Perpetual licenses were discontinued, and customers were pushed toward subscription-based bundles at significantly higher price points. For many organizations, including DCC, this translated into a dramatic and unplanned increase in infrastructure licensing costs — with renewal figures that were difficult to justify against the backdrop of no new functional gains.

Faced with this financial pressure, DCC's leadership initiated a strategic review of their virtualization platform. The outcome was a decision to migrate away from VMware entirely and adopt an open-source, enterprise-grade alternative — without procuring any new hardware, and without disrupting business operations beyond an acceptable downtime threshold of 24 hours per application.

DCC engaged **Illumia Solutions Pvt. Ltd. (ISPL)**, a Proxmox Silver Partner with deep expertise in open-source virtualization and hyper-converged infrastructure, to lead the migration end-to-end.



Problem Statement

DCC's decision to migrate away from VMware was driven by a convergence of financial, operational, and strategic challenges that made the status quo unsustainable.

Escalating Licensing Costs

The most immediate trigger was the sharp increase in VMware licensing costs following Broadcom's acquisition. The transition from perpetual to mandatory subscription-based licensing left DCC facing renewal costs that were disproportionate to the business value being delivered. For an IT services organization operating on defined margins, this level of unplanned expenditure posed a direct threat to profitability.

Vendor Lock-in and Strategic Risk

Beyond the immediate cost impact, DCC's leadership recognized a deeper strategic risk — continued dependence on a single vendor whose pricing and licensing policies were now subject to change without predictable roadmaps. Remaining on VMware meant accepting ongoing exposure to further cost escalations and policy shifts outside DCC's control.

Operational Continuity Constraints

The migration could not be treated as a simple lift-and-shift exercise. DCC's hosted applications were actively serving business functions, and any migration approach had to respect a maximum tolerable downtime of 24 hours per workload. A big-bang cutover — shutting everything down and rebuilding — was technically possible but carried unacceptable operational risk.

Hardware Investment Constraint

DCC's leadership made a clear commercial decision: the migration must be executed entirely on existing hardware. Procuring new servers or storage equipment was off the table. This placed a significant constraint on the migration strategy, requiring a phased, surgical approach that could repurpose live infrastructure without a parallel hardware environment to fall back on.

Backup Infrastructure Dependency

The existing Veeam Backup & Replication setup, while functional, was tied to the VMware ecosystem. Retaining it post-migration would mean carrying forward a licensing cost with limited justification. A replacement backup solution, natively integrated with the new platform, was required.

In summary, DCC needed a migration partner who could deliver a zero-new-hardware, minimum-downtime transition from VMware vSphere with vSAN to an enterprise-grade open-source platform — while simultaneously replacing the backup infrastructure and decommissioning the legacy environment cleanly.

Solution Proposed

ISPL proposed a phased, non-disruptive migration from VMware vSphere with vSAN to **Proxmox Virtual Environment (Proxmox VE)** with **Ceph** as the distributed storage backend, complemented by **Proxmox Backup Server (PBS)** as the replacement for Veeam. The solution was architected entirely around DCC's existing hardware fleet, with no new equipment involved.

Platform Selection — Proxmox VE with Ceph

Proxmox VE is an open-source, enterprise-grade virtualization platform built on KVM and LXC, with a mature web-based management interface and strong community and commercial support. As a Proxmox Silver Partner, ISPL brought certified expertise and direct vendor-backed credibility to the engagement.

Ceph was selected as the distributed storage layer for its functional equivalence to vSAN — providing redundant, self-healing, hyper-converged storage across cluster nodes without reliance on a proprietary storage array. The existing NVMe and SSD drives within each host were repurposed purposefully: **SSD drives were designated as Ceph OSDs** (the primary storage tier), while **NVMe drives were allocated to Ceph's DB and WAL functions** — accelerating metadata operations and write performance, extracting maximum throughput from the hardware already in place.

Phased Node Liberation Strategy

Since no new hardware was available, the core challenge was bootstrapping a Proxmox VE cluster using nodes that were still active members of the VMware vSAN cluster. ISPL's approach addressed this through a carefully sequenced node liberation strategy:

- Spare drives within existing ESXi hosts were used to create **local datastores**, providing temporary holding capacity for VM workloads
- VMs were selectively migrated from the vSAN pool to these local datastores to reduce vSAN utilization to a level where nodes could be safely evacuated
- Nodes were then withdrawn from the VMware cluster in a controlled manner — one set at a time — and re-provisioned as Proxmox VE nodes
- This cycle was repeated iteratively until all nodes had been migrated

This approach ensured that the VMware cluster remained operational and degraded gracefully, rather than being taken offline abruptly.

Minimum Downtime VM Migration

Rather than leveraging the 24-hour downtime window as a design assumption, ISPL chose to treat it as a ceiling — not a target. Proxmox VE's built-in **ESXi-to-PVE migration tool** was used to migrate virtual machines with minimal downtime, converting and transferring VMs while keeping disruption to individual workloads as brief as technically feasible.

Backup Infrastructure Replacement

Proxmox Backup Server was proposed as a direct replacement for Veeam. PBS integrates natively with Proxmox VE, supporting incremental backups, deduplication, and integrity verification. Existing storage drives from the decommissioned Veeam infrastructure were repurposed as the PBS datastore. Backup data at rest was configured with **encryption** and a defined **retention policy**, ensuring data security and storage efficiency from day one.

Implementation

ISPL executed the migration in a structured sequence of phases, each designed to maintain cluster stability, protect live workloads, and progressively transfer the infrastructure from VMware to Proxmox VE. The entire engagement was carried out on DCC's existing hardware without any new equipment.

Phase 1 — Assessment and Pre-Migration Preparation

Before any changes were made to the production environment, ISPL conducted a thorough assessment of the existing VMware cluster. This included inventorying all hosted virtual machines, understanding their resource profiles and application dependencies, and identifying workload criticality in consultation with DCC's application stakeholders. ESXi hosts carrying a comparatively lighter VM load were identified as the primary candidates for the first wave of node liberation. Spare drives available within other ESXi hosts were also identified and mapped, as these would serve as local datastore capacity to absorb workloads during the migration process.

Phase 2 — Freeing the Initial Three Nodes

ISPL selected three ESXi hosts that were running a comparatively smaller number of virtual machines — making them the most practical candidates for the first wave of liberation. The VMs running on these three hosts were migrated to other ESXi hosts remaining in the cluster. To accommodate the incoming workloads, those receiving ESXi hosts utilized spare NVMe and SSD drives to provision local datastores, providing the necessary storage capacity outside of the vSAN pool. Once all VMs had been successfully relocated, the three target ESXi hosts were completely free of workloads and ready for controlled evacuation from the VMware cluster.

Phase 3 — Controlled Node Evacuation from VMware

With the three target hosts emptied of workloads, ISPL proceeded with their formal removal from the VMware cluster. For each node, the following sequence was followed:

- The ESXi host was placed into **maintenance mode** with the vSAN data migration policy set to "**Full Data Migration**" — ensuring all data hosted on that node was redistributed across the remaining cluster members before the node was touched
- The cluster was closely monitored until all vSAN objects had completed resyncing and the cluster health returned to a fully healthy state
- Only then was the node formally removed from the vCenter (VCSA) vSphere cluster

This disciplined, one-node-at-a-time approach ensured that the VMware cluster remained stable and data-safe throughout the evacuation process.

Phase 4 — Proxmox VE Cluster Formation and Ceph Configuration

The three evacuated nodes were provisioned as a **Proxmox VE cluster**. Once the cluster was established, Ceph was configured as the distributed storage backend:

- **SSD drives** were configured as Ceph OSDs, forming the primary storage tier

- **NVMe drives** were assigned to host Ceph's **DB and WAL** components, accelerating OSD performance and write throughput
- The Ceph cluster was validated for health, redundancy, and performance before any production workloads were introduced

This three-node Proxmox VE + Ceph cluster became the initial landing zone for migrating VMs away from VMware.

Phase 5 — Iterative VM Migration and Cluster Expansion

With the Proxmox VE cluster operational, VM migration began in earnest. ISPL used Proxmox VE's built-in **ESXi-to-PVE migration tool** to convert and transfer virtual machines with minimal downtime — significantly below the 24-hour threshold agreed with DCC.

The migration followed a deliberate sequence:

- VMs residing on the local datastores of the receiving ESXi hosts were migrated to the Proxmox VE cluster first, as these carried no vSAN dependency and could be moved immediately
- With those VMs cleared, the ESXi hosts that had been temporarily hosting them were now candidates for the next round of liberation — their remaining vSAN-backed VMs were redistributed across the still-active VMware nodes to thin the pool further
- These ESXi hosts were then evacuated from VMware using the same controlled process defined in Phase 3, and joined the growing Proxmox VE + Ceph cluster
- This cycle of **migrate → evacuate → expand** continued iteratively until all nodes had been decommissioned from VMware and absorbed into the Proxmox VE cluster

Throughout this phase, application stakeholders were kept engaged, with workload health verified after each migration wave.

Phase 6 — Workload Verification and VMware Decommissioning

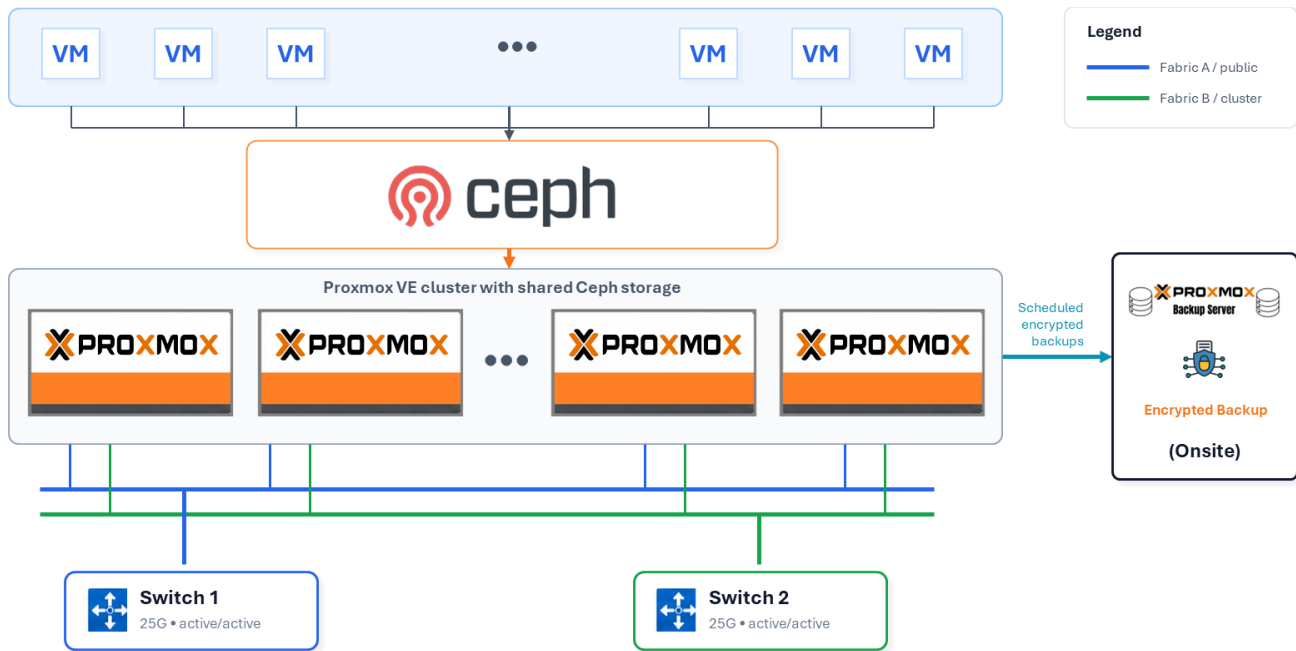
Once all ESXi nodes had been evacuated and all VMs had been migrated to the Proxmox VE cluster, a comprehensive verification exercise was conducted. Application stakeholders confirmed the operational status of each hosted workload. Only after sign-off was obtained was the VMware environment — including vCenter Server (VCSA) and all ESXi host licenses — formally decommissioned.

Phase 7 — Proxmox Backup Server Deployment and Veeam Decommissioning

With the Proxmox VE cluster fully operational and verified, attention turned to replacing the backup infrastructure. Proxmox Backup Server was deployed and the existing storage drives — previously serving the Veeam environment — were repurposed as the PBS backup datastore. Key configurations included:

- **Encryption at rest** for all backup data, ensuring data security compliance
- A defined **retention policy** governing backup frequency and data lifecycle
- Integration with the Proxmox VE cluster for scheduled, incremental VM backups with deduplication

Following successful PBS configuration and a validation of backup and restore operations, the Veeam Backup & Replication environment was decommissioned.



Conclusion

The migration of Data Consultants Corporation's virtualization infrastructure from VMware vSphere with vSAN to Proxmox VE with Ceph stands as a strong example of how strategic platform decisions, when executed with the right expertise, can deliver measurable business outcomes without compromising operational continuity.

Business Outcomes

The most immediate and tangible outcome was a **dramatic reduction in licensing costs**. By transitioning to Proxmox VE — an open-source platform with transparent, predictable licensing — DCC eliminated its exposure to Broadcom's VMware subscription model entirely. For an IT services organization where infrastructure cost directly impacts margin, this is a significant and recurring financial gain, not a one-time saving.

Operational Continuity

Despite the complexity of the migration — involving a live, production-grade hyper-converged cluster with no new hardware available — application downtime was kept well within the agreed 24-hour threshold. In practice, ISPL's use of Proxmox VE's native ESXi-to-PVE migration tooling meant that individual workloads experienced only minimal disruption, far below what the business had braced for. DCC's client-facing and internal services continued operating throughout the migration lifecycle.

No New Hardware Investment

The entire migration was executed on DCC's existing server and storage fleet. Spare drives were repurposed as local datastores and Ceph performance tiers. The decommissioned Veeam storage was reused for Proxmox Backup Server. This zero-new-hardware outcome validated the feasibility of ISPL's phased approach and delivered the migration entirely within DCC's capital expenditure constraints.

A Future-Ready Infrastructure

DCC's new infrastructure — Proxmox VE with Ceph for compute and storage, and Proxmox Backup Server for data protection — is an integrated, open-source stack with no vendor lock-in. Backup data is encrypted at rest with a defined retention policy, strengthening DCC's data security posture. The platform is well-positioned to scale as DCC's business grows, without the unpredictability of proprietary licensing decisions made by a third party.

ISPL's Role

As a Proxmox Silver Partner, ISPL brought not just technical execution capability but also platform-specific expertise that was critical to the success of this engagement. The phased node liberation strategy, the disciplined vSAN evacuation process, and the iterative migrate-evacuate-expand approach were all products of deep familiarity with both the VMware and Proxmox ecosystems — enabling a migration that was safe, efficient, and commercially sound.

In conclusion, DCC's VMware-to-Proxmox migration delivered on every dimension that mattered to its leadership: cost reduction, operational continuity, hardware reuse, and a modernized, vendor-independent infrastructure foundation — all within the project's defined constraints.